

Utkarsh Ravindrakumar Mahajan

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TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Python, C/C++, SQL, Bash

Frontend: React, React Native, Next.js, TanStack Start, Tailwind, shadcn/ui, WebSockets

Backend & Data: Node.js, Express, Django, PostgreSQL, MySQL/MariaDB, Redis, Kafka

Infrastructure & Security: Docker, Kubernetes, Linux, OCI, GitLab CI/CD, Prometheus, Grafana, OAuth 2.0 / OIDC, RBAC

EXPERIENCE

Applications Solution Developer

June 2024 – Present

Oracle India Private Limited

Hyderabad, IN

- Reduced manual document validation by **5–6K hours** annually with an AI workflow on OCI automating **60%** of checks
- Governed Fusion AI agent access by designing an A2A Gateway with rubric-based tiering, per-user RBAC, and audit logging
- Automated ML model deployment by building an MLOps pipeline on Kubernetes, Kafka, and OCI Data Science
- Cut MCP server onboarding org-wide by designing reusable container base images with OAuth 2.0 auth and self-registration
- Received **5 Rock Star Gold Awards**, Oracle's internal recognition for delivery impact

Systems Engineer

July 2022 – March 2024

IRIS, NITK Surathkal

Surathkal, IN

- Reduced attack surface on the student-run IRIS LMS (**24K+** users) via production security and config hardening
- Standardized deployments across services by automating builds with GitLab CI/CD and containerizing them
- Ensured availability during peak exam usage by conducting distributed load testing with Locust, k6, and Grafana

EDUCATION

National Institute of Technology Karnataka

2020 – 2024

Bachelor of Technology in Electronics and Communication Engineering

CGPA: 7.87/10

PROJECTS

Kneed, Knee Recovery App | React Native, TypeScript, Node.js, Express, SQLite

Live on App Store, 2025

- Shipped a solo-built iOS rehab app end to end: OAuth 2.1 API, Apple Sign-In, in-app purchases, App Store release
- Synced Apple HealthKit sensor data (steps, gait speed, sleep) into a phased exercise and pain-trend engine
- Built an MCP server exposing recovery data so an LLM can configure and adjust programs conversationally

Clapped, Learning List Platform | React 19, TanStack Start, PostgreSQL, Drizzle ORM

clapped.space, 2025

- Shipped an SSR-first React 19 app with public profiles, sharing, ratings, and progress tracking
- Composed a rich-text editor on Tiptap supporting markdown, code blocks, math rendering, and Excalidraw embeds
- Enabled conflict-free drag-and-drop reordering using fractional indexing instead of integer position rewrites

Execstasy, Lightweight IAM for Linux | Go, C, Next.js, React, PostgreSQL, Redis

2025

- Designed a React and shadcn/ui console for SSH access requests, approvals, and permission management
- Eliminated per-instance credentials and manual key distribution by centralizing access behind a Go RBAC API
- Enabled password-less SSH by writing a Linux PAM module in C implementing OAuth 2.0 Device Flow (RFC 8628)

Staging Server | Django, Celery, Redis, Docker, Nginx, WebSockets

IRIS-NITK, 2023 – 2024

- Gave every IRIS-NITK developer one-click deploys by building a staging platform for containerized apps
- Removed server-access needs when debugging by adding WebSocket log streaming and an in-browser xterm.js terminal

MySQL/MariaDB eBPF Query Profiler | C, Go, eBPF, Prometheus, Grafana

2023

- Surfaced per-query latency in real time through fingerprinted Prometheus metrics and Grafana dashboards
- Measured **18%** higher throughput than Performance Schema on 200ms+ queries (sysbench OLTP) using eBPF

ACHIEVEMENTS & COMMUNITY

- Contributed to KubeArmor (CNCF), Vitess, Kerbrute, and GoKrb5; authored **20+** technical write-ups
- Earned Hack The Box CPTS, an OSCP-equivalent hands-on penetration testing certification
- Placed **27th globally** at NahamCon EU CTF 2022 and competed in several CTFs since